

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 9 (M2)

AIM : Conducting Chi-square tests using chisq.test().

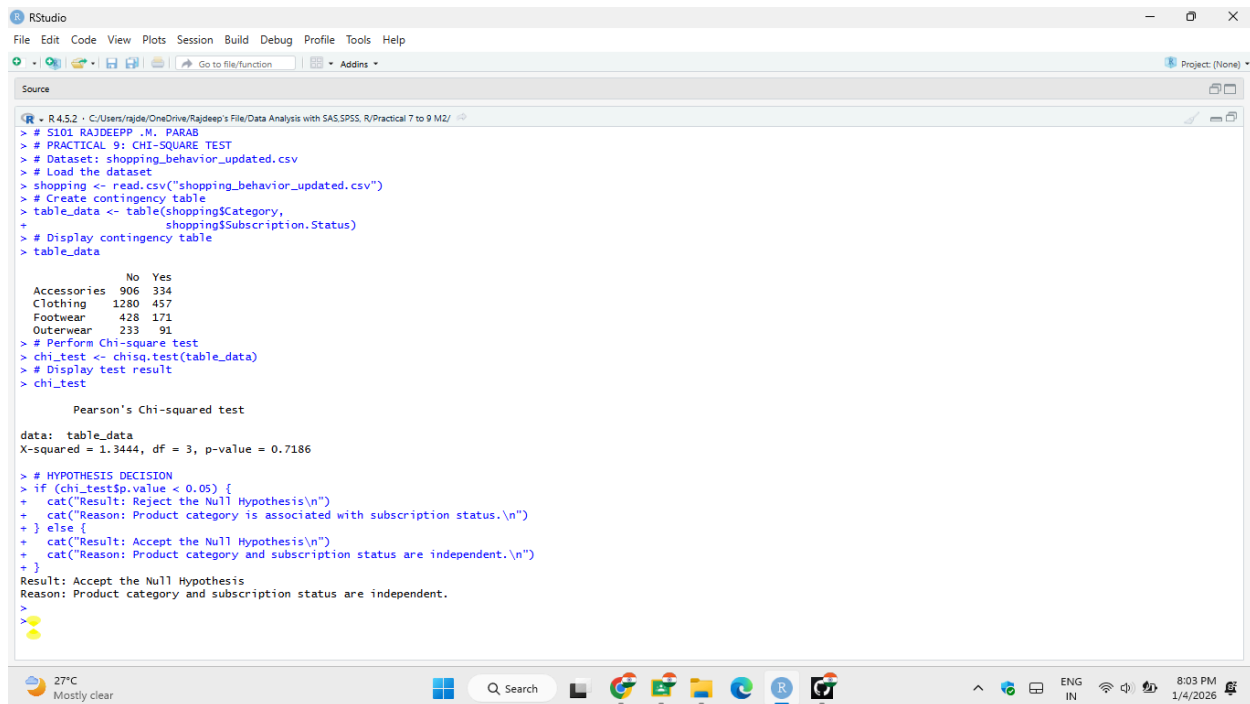
Input :

```
# S101 RAJDEEPP .M. PARAB
# PRACTICAL 9: CHI-SQUARE TEST
# Dataset: shopping_behavior_updated.csv
# Load the dataset
shopping <- read.csv("shopping_behavior_updated.csv")
# Create contingency table
table_data <- table(shopping$Category,
                    shopping$Subscription.Status)
# Display contingency table
table_data
# Perform Chi-square test
chi_test <- chisq.test(table_data)
# Display test result
chi_test
# HYPOTHESIS DECISION
if (chi_test$p.value < 0.05) {
  cat("Result: Reject the Null Hypothesis\n")
  cat("Reason: Product category is associated with subscription status.\n")
} else {
  cat("Result: Accept the Null Hypothesis\n")
  cat("Reason: Product category and subscription status are independent.\n")
}
```

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Output :



```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
R - R 4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/
> # S101 RAJDEEPP .M. PARAB
> # PRACTICAL 9: CHI-SQUARE TEST
> Dataset: shopping_behavior_updated.csv
> # Load the dataset
> shopping <- read.csv("shopping_behavior_updated.csv")
> # Create contingency table
> table_data <- table(shopping$Category,
+                     shopping$Subscription.Status)
> # Display contingency table
> table_data

      No  Yes
Accessories 906 334
Clothing 1280 457
Footwear 428 171
Outerwear 233 91
> # Perform Chi-square test
> chi_test <- chisq.test(table_data)
> # Display test result
> chi_test

      Pearson's Chi-squared test

data:  table_data
X-squared = 1.3444, df = 3, p-value = 0.7186

> # HYPOTHESIS DECISION
> if (chi_test$p.value < 0.05) {
+   cat("Result: Reject the Null Hypothesis\n")
+   cat("Reason: Product category is associated with subscription status.\n")
+ } else {
+   cat("Result: Accept the Null Hypothesis\n")
+   cat("Reason: Product category and subscription status are independent.\n")
+ }
Result: Accept the Null Hypothesis
Reason: Product category and subscription status are independent.
>
```